

ITA0607-Machine Learning

CO1-AT1

Bandaru Jayasri

192424301

10. Quality Inspection System

Given:

Actual / Predicted

	Defective	Non-Defective
Defective	230 (TP)	20 (FN)
Non-Defective	25 (FP)	225 (TN)

Total Products = 500

- TP = 230
- TN = 225
- FP = 25
- FN = 20

(i) Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$
$$\text{Accuracy} = \frac{230 + 225}{230 + 225 + 25 + 20} = \frac{455}{500} = 0.91$$

Accuracy = 91%

(ii) Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$
$$\text{Precision} = \frac{230}{230 + 25} = \frac{230}{255} = 0.90196$$

Precision = 90.20%

(iii) Sensitivity (Recall)

$$\text{Sensitivity} = \frac{TP}{TP+FN} = \frac{230}{230+20} = \frac{230}{250} = 0.92$$

Sensitivity = 92%

(iv) Specificity

$$\text{Specificity} = \frac{TN}{TN+FP} = \frac{225}{225+25} = \frac{225}{250} = 0.90$$

Specificity = 90%

(v) F1-Score

$$F1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} = \frac{2 \times (0.90196 \times 0.92)}{0.90196 + 0.92} = \frac{1.6596}{1.82196} = 0.9109$$

F1-Score = 91.09%

Final Answers

Metric	Value
Accuracy	91%
Precision	90.20%
Sensitivity	92%
Specificity	90%
F1-Score	91.09%

11. Shop X and Shop Y

Find the probability that the bottle came from **Shop Y**, given that it is adulterated.

Given

Shop X:

- Pure = 20
- Adulterated = 30
- Total = 50

Shop Y:

- Pure = 40
- Adulterated = 20
- Total = 60

Shop selection probabilities:

$$P(X)=\frac{1}{3}, P(Y)=\frac{2}{3} \quad P(X)=\frac{1}{3}, P(Y)=\frac{2}{3}$$

Probability of adulterated bottle:

$$P(A|X)=\frac{30}{50}=0.6 \quad P(A|X)=\frac{30}{50}=0.6$$

$$P(A|Y)=\frac{20}{60}=\frac{1}{3} \quad P(A|Y)=\frac{20}{60}=\frac{1}{3}$$

Using **Bayes' Theorem**:

$$P(Y|A)=\frac{P(A|Y)P(Y)}{P(A|X)P(X)+P(A|Y)P(Y)} \quad P(Y|A)=\frac{P(A|Y)P(Y)}{P(A|X)P(X)+P(A|Y)P(Y)}$$

Substitute values:

$$=\frac{(1/3)(1/3)}{(1/3)(1/3)+(1/3)(1/3)} = \frac{(1/9)}{(1/9)+(1/9)} = \frac{1/9}{2/9} = \frac{1}{2}$$

$$LCM = 45$$

$$=\frac{10/45}{10/45+10/45} = \frac{10}{10+10} = \frac{10}{20} = \frac{1}{2} = 0.5$$

Final Answer

$$P(\text{Shop Y} | \text{Adulterated}) = \frac{1}{2} \approx 50\%$$

12. Warehouse Sections

Find the probability that the defective item came from **Section A**.

Given

Section A:

- Defective = 15
- Non-defective = 35
- Total = 50

Section B:

- Defective = 25
- Non-defective = 25
- Total = 50

Section selection probabilities:

Section A is three times as likely as Section B.

$$P(A)=\frac{3}{4}, P(B)=\frac{1}{4} \quad P(A)=\frac{3}{4}, P(B)=\frac{1}{4}$$

Defective probabilities:

$$P(D|A)=\frac{15}{50}=0.3 \quad P(D|A)=\frac{15}{50}=0.3 \\ P(D|B)=\frac{25}{50}=0.5 \quad P(D|B)=\frac{25}{50}=0.5$$

Using Bayes' Theorem:

$$P(A|D)=\frac{P(D|A)P(A)}{P(D|A)P(A)+P(D|B)P(B)} \quad P(A|D)=\frac{P(D|A)P(A)}{P(D|A)P(A)+P(D|B)P(B)}$$

Substitute values:

$$= \frac{0.3 \times 0.75}{0.3 \times 0.75 + 0.5 \times 0.25} = \frac{0.225}{0.225 + 0.125} = \frac{0.225}{0.35} = 0.6429 \approx 64.29\%$$

Final Answer

$$P(\text{Section A} | \text{Defective}) = 0.6429 \approx 64.29\% \quad P(\text{Section A} | \text{Defective}) = 0.6429 \approx 64.29\%$$

Final Answers Summary

Question	Answer
10	Accuracy = 91% , Precision = 90.20% , Sensitivity = 92% , Specificity = 90% , F1-Score = 91.09%
11	$P(\text{Shop Y} \text{Adulterated}) = \frac{10}{19} \approx \mathbf{52.63\%}$ $P(\text{Shop Y} \text{Adulterated}) = \frac{19}{10} \approx \mathbf{52.63\%}$
12	$P(\text{Section A} \text{Defective}) = \mathbf{64.29\%}$ $P(\text{Section A} \text{Defective}) = \mathbf{64.29\%}$